UNIT 13

INFORMATION HANDLING

DATA

"Data is set of information and facts which is collected in the form of figures by observation and measurement."

We can collect data by several ways i.e. observation, survey, interview, questionnaire etc.

TYPES OF DATA

There are two types of data.
1. Ungrouped data
2. Grouped data

1. Ungrouped data:
Data which shows the individual information is known as ungrouped data.

Example:
Following are the marks scored out of 100 by 15 students of class VI in test of Science.
76, 51, 65, 62, 70, 63, 69, 62, 75, 61, 80, 55, 65, 59, 60.

2. Grouped data:
If we arrange or classify the data in groups, it is called grouped data.

Example:
In the above example of ungrouped data, student marks range from 50 to 80.
The marks can be organised into following groups.
51-60, 61-70, 71-80

Marks	Marks Obtained	No. of Students
51 - 60	51, 55, 59, 60	4
61 - 70	61, 62, 62, 63, 65, 65, 69, 70	8
71 - 80	75, 76, 80	3

EXERCISES 13.1

Q1. Which table is showing a grouped data?

(i) Total marks secured by 12 students in an examination.

Names of Student	Ali	Sana	Adil	Amir	Asma	Huma	Arif	Babar	Anis	Anil	Umar	Sara
Marks	581	786	725	788	580	690	780	599	509	619	560	678

(ii) Raza Book Store's sale in a month is shown by the table:

Number of books sold	Days
512 - 611	8
612 - 711	5
712 - 811	4
812 - 911	9

(iii) Visitors in Sindh Museum.

Number of visitors	Days
150 - 199	3
200 - 249	4
250 - 299	1
300 - 349	2
350 - 399	3
400 - 499	2

(iv) Runs scored by players of our school cricket team in one day match.

Player	1	2	3	4	5	6	7	8	9	10	11
Runs	62	41	15	59	22	10	8	2	43	7	21

Q2. Look at the chart given below and answer the given questions.

Items	How many children in the school have it
Pens	47
Pencils	62
Crayon	51
Eraser	48
Pencil Box	24

1. How many children have crayon.
2. How many children have pencils?
3. Which item is the least that children have?

Q3. Look at the chart given below and answer the given questions.

Apple Trees	Mango Trees	Pear Trees	Orange Trees
15	25	10	35

1) How many apple and mango trees are there in the orchard?

____

2) How many trees are there in all? ___________

3) Which type of tree is of least number in the orchard?

____

4) How many more orange trees are there than pear trees?

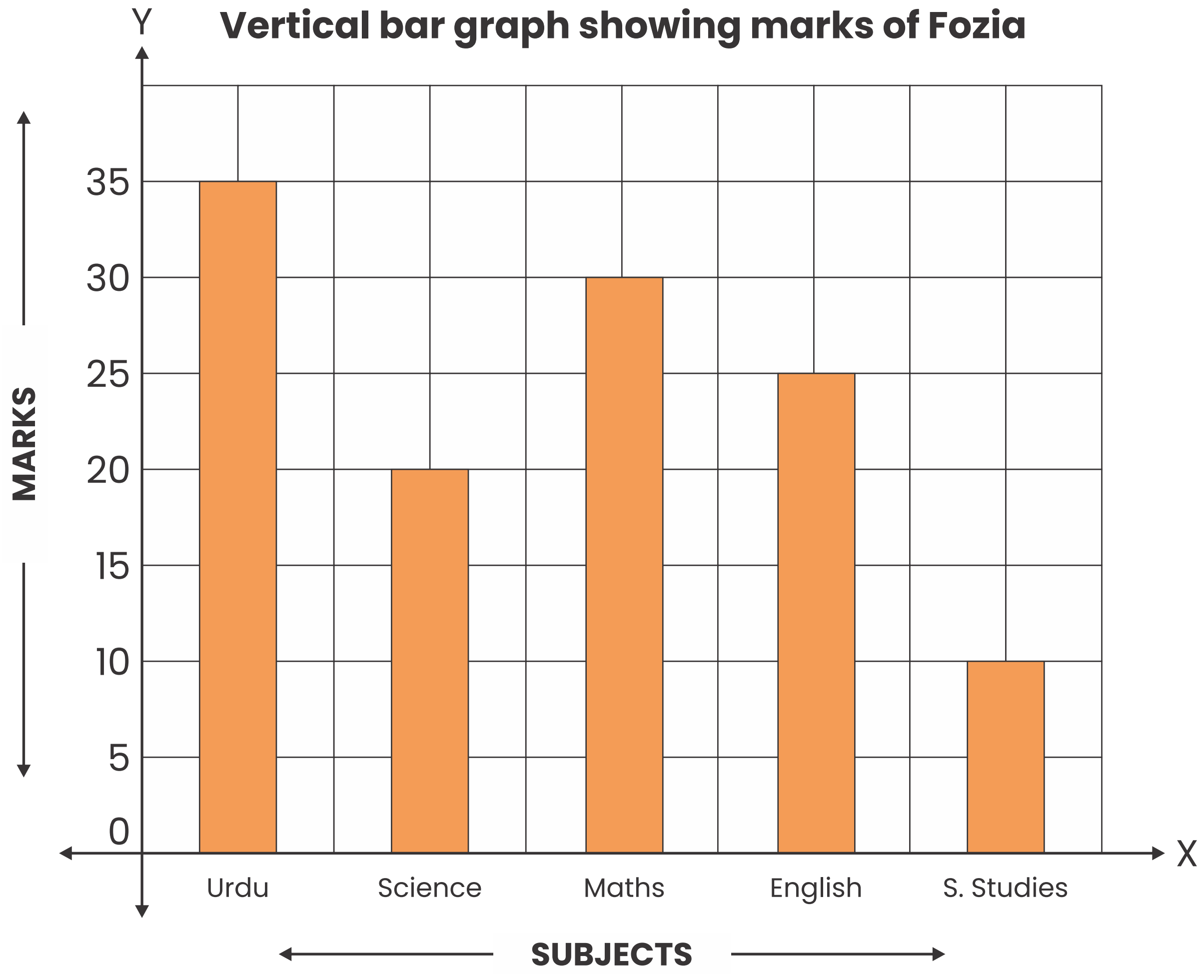
____

BAR GRAPH

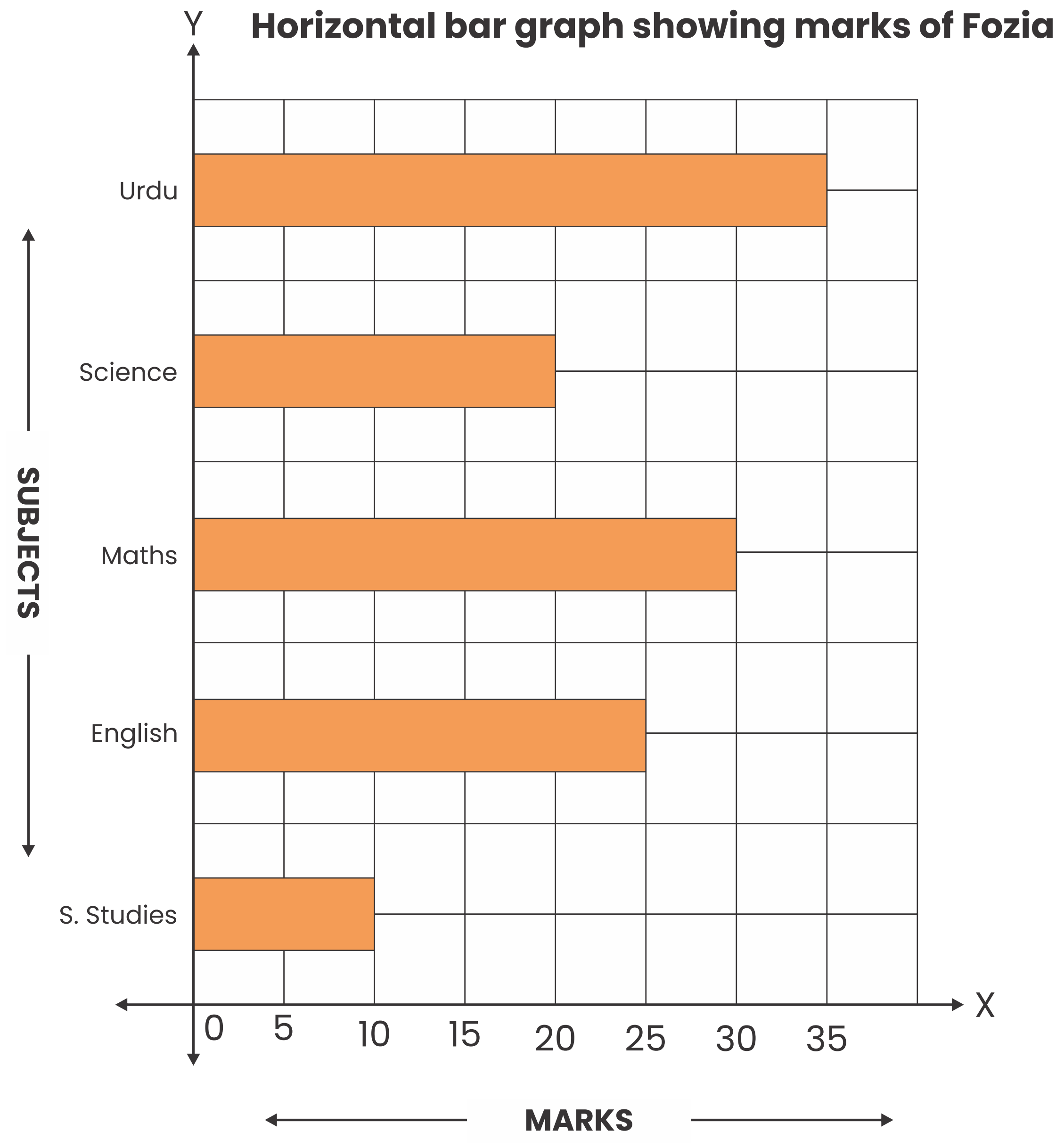
In bar graph data is represented by a number of rectangular bars of the same width with equal spacing.
There are two ways to draw bar graph.

1. Vertical Bar Graph
2. Horizontal Bar Graph

1. Vertical Bar Graph



2. Horizontal Bar Graph



Steps to draw horizontal and vertical bar graphs.

Step 1.

Take a graph paper and draw two lines **OX** and **OY**, perpendicular to one another. **OX** is called X-axis and **OY** is called Y-axis and **O** is the origin. Give the graph a title.

Step 2.

Decide a scale for each axis.

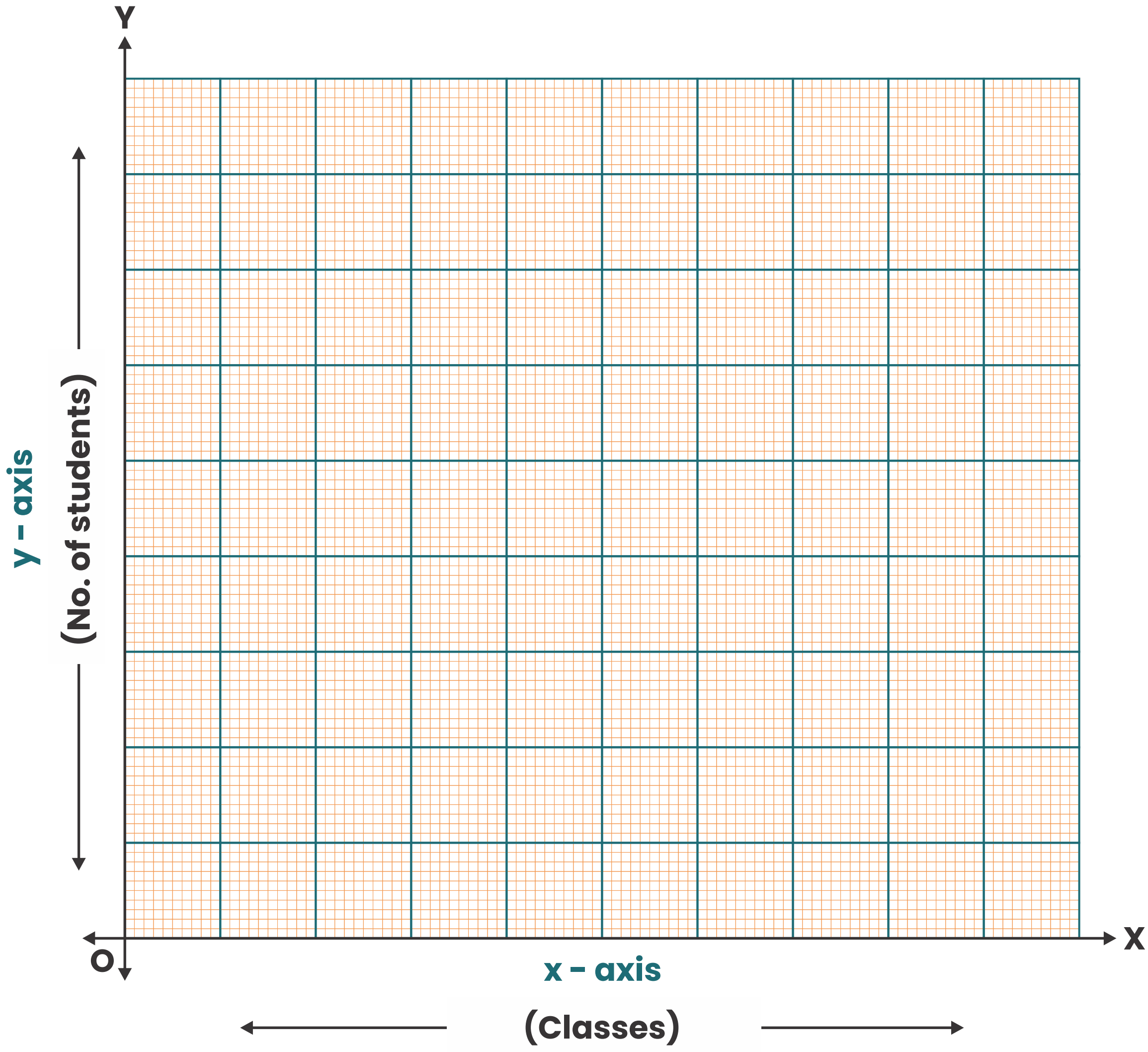
Step 3.

Make bars of equal breadth at equal distances according to the information.

Example: Draw a vertical bar graph showing the following information.

Class	I	II	III	IV	V
Number of Students	45	30	25	36	45

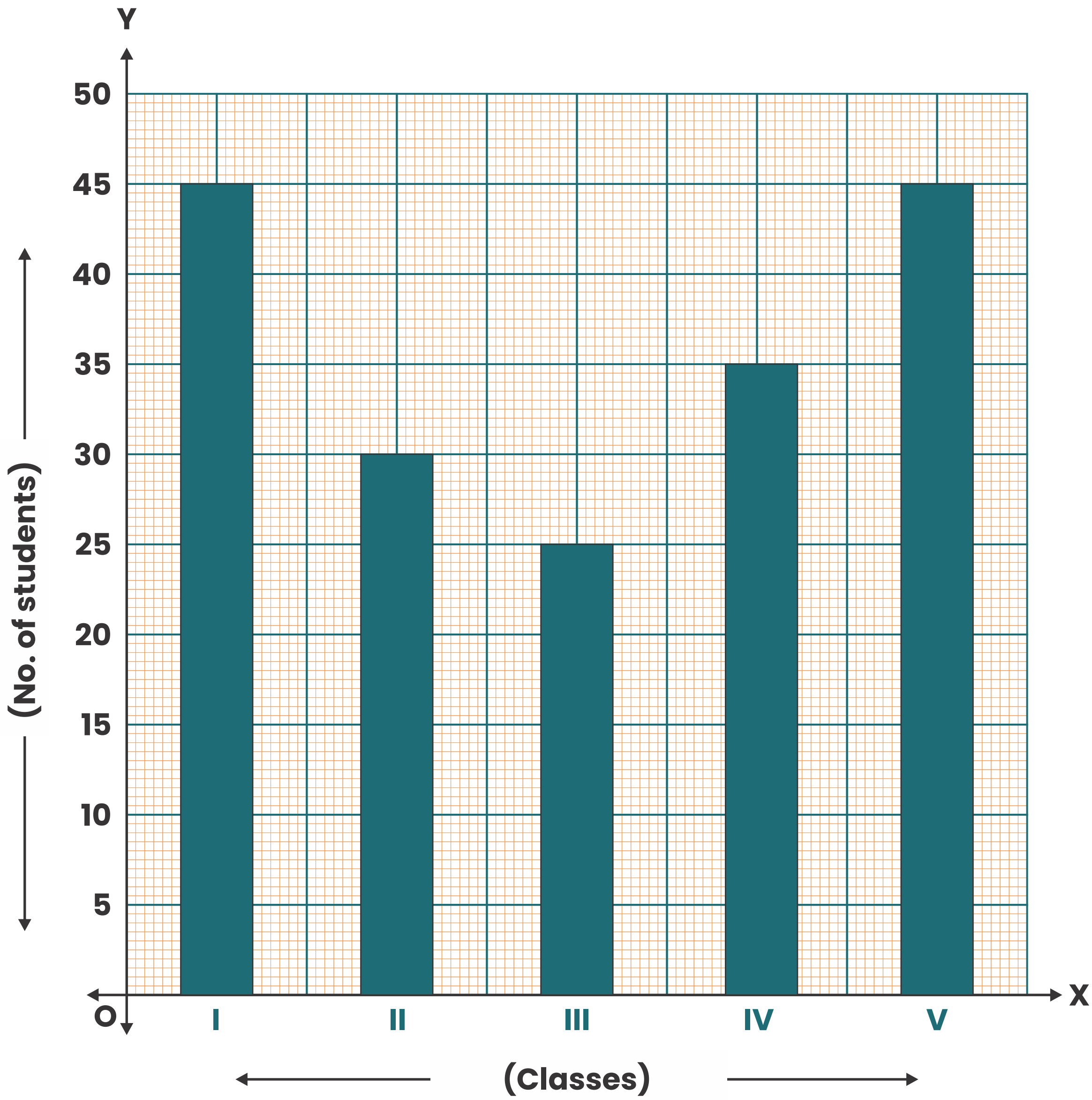
Step I: In this graph the "number of students" are shown on Y-axis and classes are shown on X-axis.



Step II:
Scale: Two small square on Y-axis represents one student.
Length of each bar is taken according to the number of students.
Breadth of each bar is 10 small squares and distance between them is 10 small squares on X-axis.

Step III:
In Class I: There are 45 students. Every two small squares on Y-axis represents a student, so we draw the bar graph up to 90 small squares to represent number of student. Similarly, draw bars for representing other classes up to the required heights.

Vertical Bar Graph representing number of students in classes I to V.

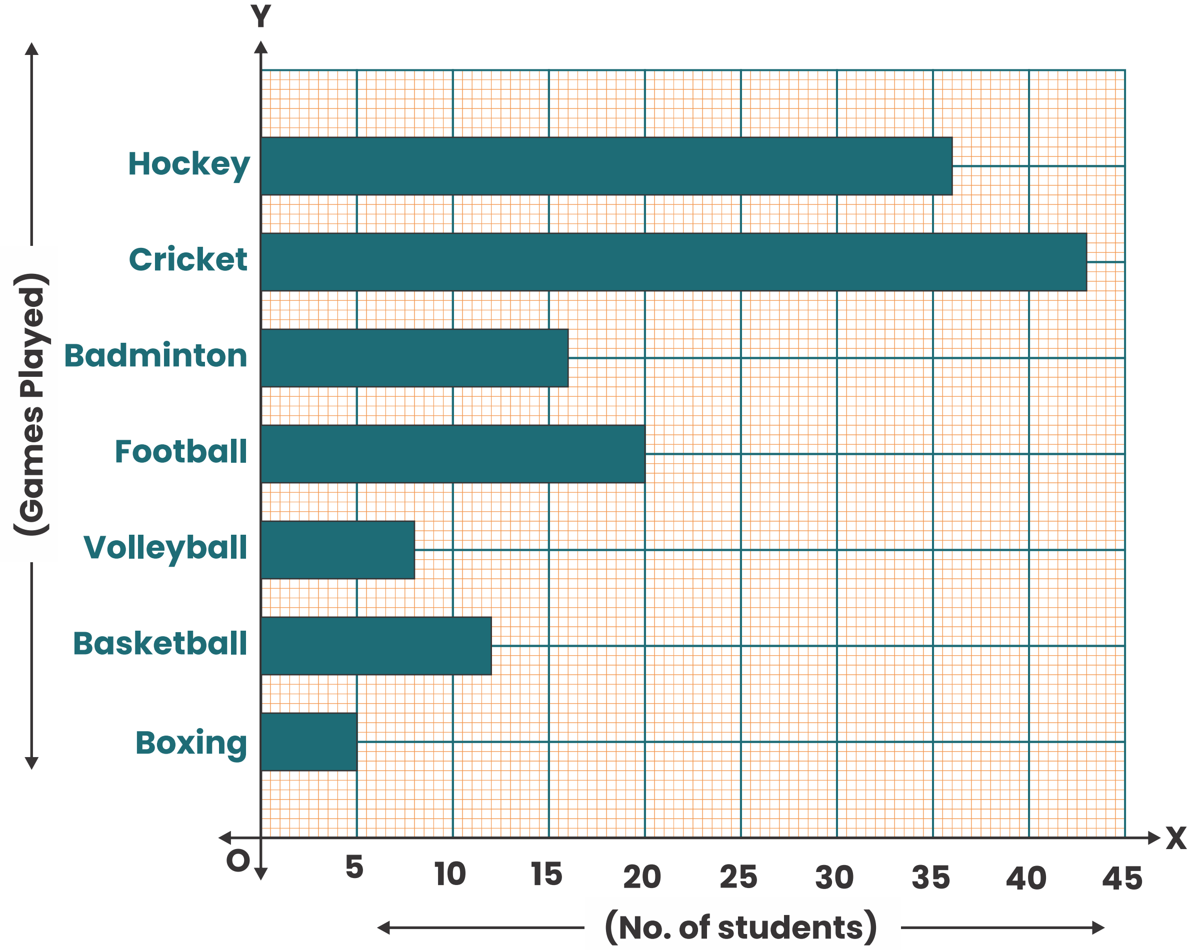


Check list for drawing the bar graph.
1. Read the information carefully.
2. Select proper axes for given information.
3. Make a scale.
4. Draw the bar for each information.
5. Label the graph and give a title.

Example: Games played by students of Class V in a school are given in the table below. Represent data in horizontal bar graph.

1. Read the table, there are seven different types of games played by students.
2. We select x-axis for number of students and y-axis for games played.
3. Make a scale for number of students and played games. 1 unit represents 2 students.
4. Draw horizontal bar for each game.
5. Label both axis and give the graph a title.

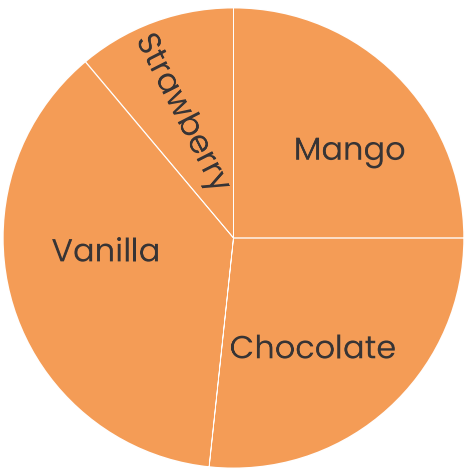
Game Played	No. of Students
Hockey	36
Cricket	43
Badminton	16
Football	20
Volleyball	8
Basketball	12
Boxing	5



PIE GRAPH

A pie chart is circle graph. When we have many quantities in data, we use a pie graph or pie chart to represent it.

Example 1: The data of cone ice cream of different flavours sold at a shop in a day are shown in the pie graph as under.

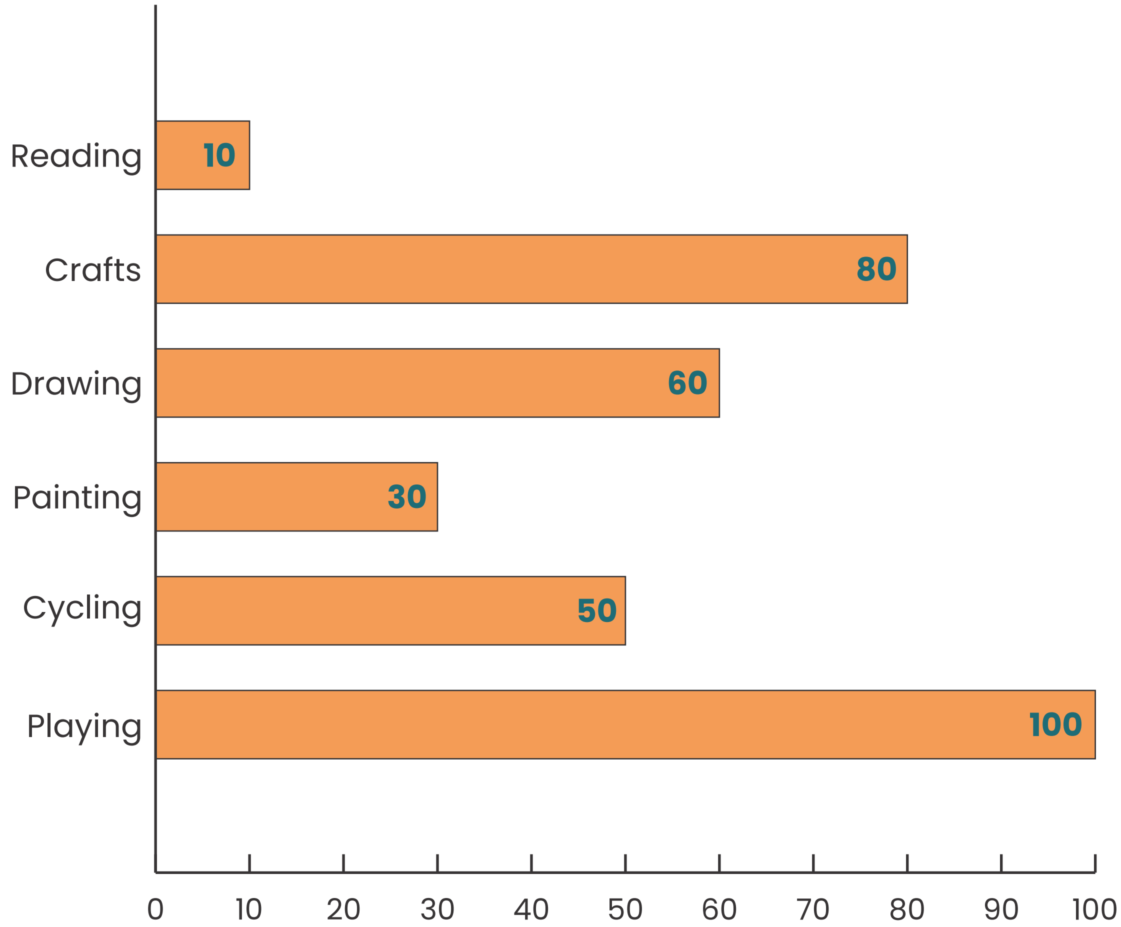


Study the graph and answer the following questions:

(i) Name the flavour with most sale.	Vanilla
How do you come to know?	Because the representing sector is largest in size.
(ii) Name the flavour with least sale.	Strawberry
How do you come to know?	Because the representing sector is smallest in size.

EXERCISES 13.2

Q1. A survey was done to find out which activity is most popular amongst children in a class and here are the findings.



1. Which activity is the most popular? ____________

2.Which activity is the least popular? ____________

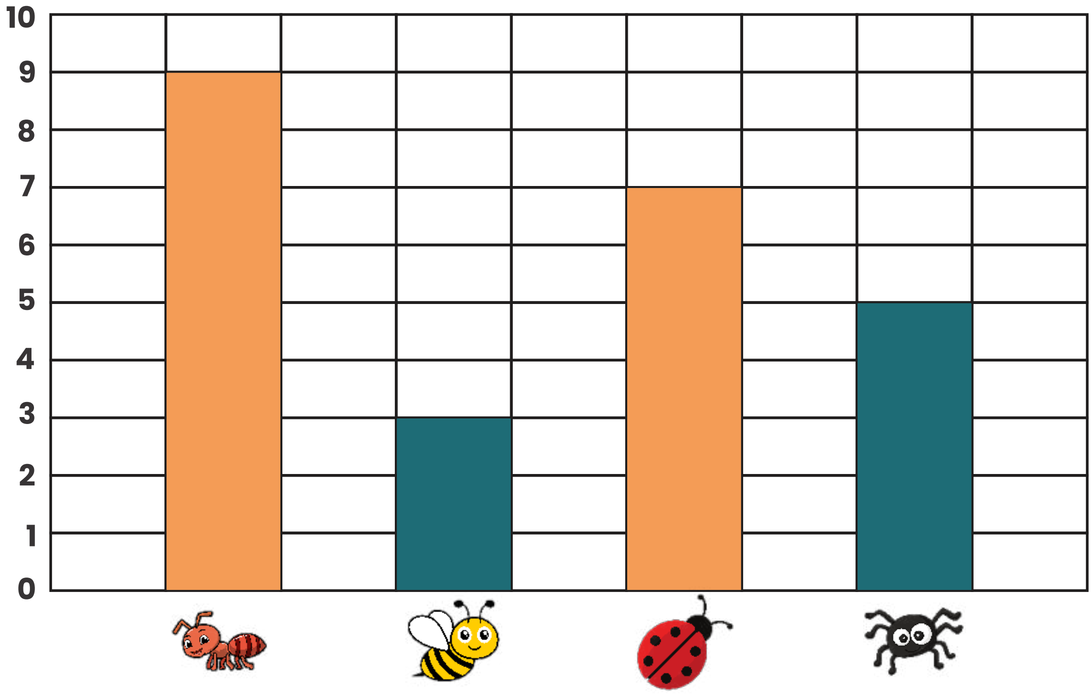
3. Which activity is the second least popular? ____________

4. How many children liked Drawing? ____________

5. How many children liked Writing? ____________

Q2. Read the bar graph and answer the questions.

Insects caught in the net



How many _______ were caught in the net?

(1)  (2) 

(3)  (4) 

(5) Which insect was caught the most?

(6) Which insect was caught the least?

(7) How many bees and spiders were caught?

Q3. Read the data of animals spotted in the garden and draw the bar graph.

Snail	Lizard	Rat	Chameleon	Frog
				
8	14	12	4	18

Q4. Draw a vertical bar graph to represent the information given below:

Monthly Sale of Fans in a Shop

Name of Months	Jan	Feb	Mar	Apr	May	June
No. of Fans	70	50	35	60	90	95

Q5. Represent the following information in a vertical bar graph.

Monthly Expenses of Akbar's Family

Items	Food	Education	Rent	Telephone	Electricity	Gas
Expenses (in rupees)	6000	5000	4500	1500	2000	300

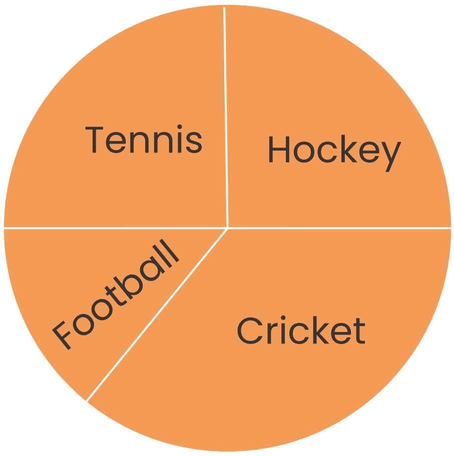
Q6. Following pie graph is representing the games played by the students of class VI.

Study the pie graph and answer the following questions.

(i) Which game is played less?

(ii) Which game is played most?

(iii) Which games are played equally?



Q7. The marks obtained in different subjects by Aftab are shown by the following pie graph.

Study the pie graph and answer the following questions.

(i) In which subject, Aftab secured highest marks?

(ii) In which subject, Aftab secured lowest marks?

(iii) In which subject, Aftab secured more marks in science or math?